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1 --Q1. What is the total revenue generated by male vs. female customers?
2
3 select gender, SUM(purchase_amount) as revenue
4 from customer
5 group by gender
6
7
8 --Q2. Which customers used a discount but still spent more than the average purchase amount?
9
10 select customer_id, purchase_amount
11 from customer
12 where discount_applied = 'Yes' and purchase_amount >= (select AVG
    (purchase_amount) from customer)
13
14
15 -- Q3. Which are the top 5 products with the highest average review rating?
16
17 select item_purchased, round(avg(review_rating::numeric),2) as "Average
    Product Rating"
18 from customer
19 group by item_purchased
20 order by avg(review_rating) desc
21 limit 5
22
23 --Q4. Compare the average Purchase Amounts between Standard and Express
    Shipping.
24
25 select shipping_type,
26 ROUND(AVG(purchase_amount),2)
27 from customer
28 where shipping_type in ('Standard', 'Express')
29 group by shipping_type;
30
31 --Q5. Do subscribed customers spend more? Compare average spend and total
    revenue
32 --between subscribers and non-subscribers.
33
34 SELECT subscription_status,
35         COUNT(customer_id) AS total_customers,
36         ROUND(AVG(purchase_amount),2) AS avg_spend,
37         ROUND(SUM(purchase_amount),2) AS total_revenue
38 FROM customer
39 GROUP BY subscription_status
40 ORDER BY total_revenue,avg_spend DESC;
41
42 --Q6. Which 5 products have the highest percentage of purchases with
    discounts applied?
43
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44 SELECT item_purchased,
45         ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0
46                               END)/COUNT(*),2) AS discount_rate
47 FROM customer
48 GROUP BY item_purchased
49 ORDER BY discount_rate DESC
50 LIMIT 5;
51
52 --Q7. Segment customers into New, Returning, and Loyal based on their
53     total
54
55 -- number of previous purchases, and show the count of each segment.
56
57 with customer_type as (
58 SELECT customer_id, previous_purchases,
59 CASE
60     WHEN previous_purchases = 1 THEN 'New'
61     WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
62     ELSE 'Loyal'
63 END AS customer_segment
64 FROM customer)
65
66 select customer_segment, count(*) AS "Number of Customers"
67 from customer_type
68 group by customer_segment;
69
70 --Q8. What are the top 3 most purchased products within each category?
71
72 WITH item_counts AS (
73     SELECT category,
74            item_purchased,
75            COUNT(customer_id) AS total_orders,
76            ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT
77                                (customer_id) DESC) AS item_rank
78     FROM customer
79     GROUP BY category, item_purchased
80 )
81
82 SELECT item_rank, category, item_purchased, total_orders
83 FROM item_counts
84 WHERE item_rank <=3;
85
86 --Q9. Are customers who are repeat buyers (more than 5 previous purchases)
87     also likely to subscribe?
88
89 SELECT subscription_status,
90        COUNT(customer_id) AS repeat_buyers
91 FROM customer
92 WHERE previous_purchases > 5
93 GROUP BY subscription_status;
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89
90 --Q10. What is the revenue contribution of each age group?
91
92 SELECT
93     age_group,
94     SUM(purchase_amount) AS total_revenue
95 FROM customer
96 GROUP BY age_group
97 ORDER BY total_revenue desc;
```